



Exploring the potential of postmastectomy radiotherapy in cN + and ypN0 breast cancer patients following neoadjuvant chemotherapy

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Abstract

Purpose This study aims to determine the potential association of postmastectomy radiotherapy and survival in patients with clinically node-positive axillary breast cancer who achieved ypN0 after neoadjuvant chemotherapy.

Methods We conducted a retrospective cohort study using the National Cancer Database. Eligible patients were women aged 18–80 with cT1–2, cN+ invasive breast cancer and achieving ypN0 status. Inverse probability weighting (IPW)-based analyses were used to assess the differences in overall survival (OS) between the radiotherapy and no radiotherapy groups. Absolute 5-year OS rates between the two groups were analyzed by nonparametric sliding-window subpopulation treatment effect pattern plot (STEPP) analysis. Sensitivity analyses were conducted using multivariable Cox regressions.

Results We identified 3,351 patients who met eligibility criteria, of whom 57.0% ($N=1,910$) did not receive radiotherapy and 43.0% ($N=1,441$) did receive radiotherapy. No significant differences in OS were observed between the radiotherapy and non-radiotherapy groups after adjustment (hazard ratio = 1.01, $P = .96$). The STEPP analysis demonstrated no significant differences between the RT and no RT groups, regardless of the composite risk. Subgroup analyses showed that the difference in OS rates between the two groups was significantly correlated with cN category, and the advantage associated with radiotherapy receipt was only observed in cN2–3 patients but not in cN1 patients (hazard ratio = 0.55; P -interaction = 0.024).

Conclusions This study further supports the NSABP B51 findings, suggesting that omitting postmastectomy radiotherapy may be reasonable for cT1–2N1M0 patients who achieved ypN0. However, postmastectomy radiotherapy may be a consideration for patients with more advanced nodal disease (cN2–3) who achieve ypN0 status after NAC, indicating that a more tailored approach may be warranted.

Keywords Breast cancer · Postmastectomy radiotherapy · Survival benefit · Inverse probability weighting · Neoadjuvant chemotherapy

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Introduction

Several randomized clinical trials and meta-analyses have established that adjuvant postmastectomy radiotherapy (PMRT) improves overall survival (OS) in patients with stage II-III breast cancer [1–4]. Two randomized clinical trials have verified that regional nodal irradiation (RNI) significantly improves disease-free survival (DFS) in patients with N1 or high-risk node-negative breast cancer [5, 6]. Furthermore, a recent individual patient data meta-analysis by EBCTCG indicated that RNI can reduce breast cancer mortality and all-cause mortality in trials conducted after the 1980s [7].

However, the use of neoadjuvant chemotherapy (NAC) prior to mastectomy has created substantial controversy regarding the identification of patient subgroups that might benefit from PMRT. NAC can reduce the size and extent of locally advanced breast cancer. Pathological complete response (pCR) has been recognized as a surrogate endpoint for predicting long-term clinical outcomes, such as DFS and OS [8]. In modern systemic treatment settings, the axillary pCR rate varies between 13 and 60% across different breast cancer subtypes with clinically positive nodes [9]. Although the EBCTCG meta-analyses showed that PMRT did not benefit patients with pathological N0 (pN0) disease at initial surgery [10], omitting PMRT by extrapolation in clinical node positive (cN+) and ypN0 patients after NAC and mastectomy remains controversial.

Combined analysis of NSABP B18 and B27 revealed that T1-3N1 patients achieving pCR after NAC had less than a 10% local-regional recurrence (LRR) rate, suggesting that radiotherapy may result in a negligible absolute reduction in the long-term risk of breast cancer mortality [11]. Many retrospective studies have demonstrated that PMRT does not reduce the risk of distant metastasis or OS in cN+ patients achieving pCR after NAC [12–16]. However, McGuire et al. reported that PMRT did reduce the risk of LRR and improved the disease-specific survival and OS in stage III (especially cN2-N3 involvement) breast cancer [12]. Similarly, Rusthoven et al. found that PMRT was independently associated with improved OS in clinically T1-3N1M0 and ypN0 patients after NAC and mastectomy, based on a National Cancer Database (NCDB) analysis from 2003 to 2011 [17].

NSABP B51/RTOG 1304 aimed to test whether radiotherapy improves the breast cancer recurrence-free interval rate in women with biopsy-proven clinical T1-3N1 disease before NAC and then become pathologically-node negative at the time of surgery. The study reported a 5-year estimated invasive recurrence-free interval rate of 91.8% in the non-RT group and 92.7% in the RT group, with no statistically significant difference observed [18, 19]. This study

demonstrated that RNI did not notably improve oncologic outcomes in the NAC setting. However, the potential benefit of PMRT for patients initially with clinical N2-N3 disease who achieve ypN0 status after NAC remains uncertain. To this end, we retrospectively analyzed a large NCDB cohort to assess whether PMRT is associated with survival in breast cancer patients with clinically node-positive axillary disease who convert to ypN0 after NAC.

Methods

Study cohort

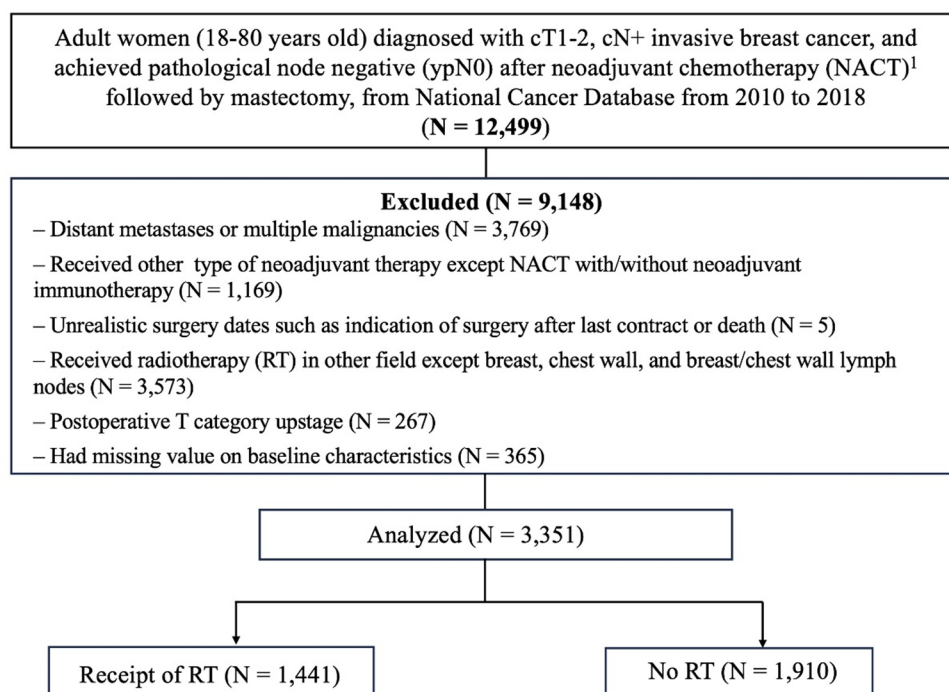
The NCDB is a hospital-based cancer registry that includes data from more than 1500 accredited facilities and represents over 70% of newly diagnosed malignancies and ~80% of all breast malignancies in the United States [20]. The NCDB was queried for eligible patients who were (1) women aged from 18 to 80; (2) diagnosed with cT1-2, cN+ invasive breast cancer between 2010 and 2018; (3) pathological node negative (ypN0) after NAC and mastectomy. The exclusion criteria included: (1) patients with distant metastases or multiple malignancies; (2) patients with post-operative T category upstaging; (3) patients who received other types of neoadjuvant therapy except NAC ± neoadjuvant anti-ERBB2 therapy¹; (4) received radiotherapy to anatomical sites other than the breast, chest wall, and regional lymph nodes (i.e., non-PMRT radiation); (5) patients who had missing values on baseline characteristics, or had unrealistic surgery dates such as indication of surgery after last contact or death. Details of patient selection are listed in Fig. 1.

Study variables and outcomes

The baseline clinicopathological variables of interest were extracted, including age at diagnosis, ethnicity (non-Hispanic White, non-Hispanic Black, Hispanic, non-Hispanic other), pathological T category, grade, hormone receptor (HR) status, ERBB2 status, histopathological classification, Charlson-Deyo score (0, 1, 2+), diagnosis year, clinical T and N category. When estrogen receptor (ER) status, and/or progesterone receptor (PR) status were positive, HR was defined as positive, whereas both ER and PR status being negative was defined as HR-negative. Of note, anti-ERBB2 directed therapy is included as immunotherapy in

¹ In NCDB, anti-ERBB2 therapy was recorded as chemotherapy before 2013 and was reclassified as BRM/immunotherapy after 2013. Therefore, we excluded patients treated with neoadjuvant BRM/immunotherapy or hormone therapy before 2013. We also excluded patients treated with neoadjuvant hormone therapy after 2013.

Fig. 1 Consolidated standards of reporting trials (CONSORT) diagram



the NCDB, and prior to 2013, it was classified as a type of chemotherapy.

Statistical analysis

Baseline characteristics were compared between patients who did and did not receive radiotherapy. The balance in baseline characteristics was evaluated by the standardized mean difference (SMD), and a threshold of 0.1 was used to indicate an acceptable balance of the factors between the two groups.

To control for confounding, we implemented inverse probability weighting (IPW) to adjust for the observed differences in baseline characteristics. Based on a multivariable logistic regression incorporating all eleven potential confounders as shown in Table 1, we first estimated the propensity scores (PSs), yielding the conditional probability of receiving radiotherapy. We then conducted IPW using estimated PS to weigh each patient and for balancing baseline characteristics across the two groups (no radiotherapy vs. radiotherapy).

The primary outcome of interest was OS, defined as the time interval between the date of initial diagnosis and the date of last contact or death. We analyzed OS by investigating IPW-adjusted Kaplan-Meier (KM) curves and log-rank tests, as well as IPW-adjusted Cox proportional hazards models. In addition, we performed subgroup IPW-adjusted analyses to test for differences in OS between patients with and without radiotherapy according to age, ethnicity (non-Hispanic White and non-Hispanic Black), pT category,

HR status, ERBB2 status, grade (grade I-II and grade III), molecular subtype (triple-negative breast cancer, HR-positive/ERBB2-negative breast cancer, and ERBB2-positive breast cancer), and cN category (cN1 and cN2-3) following the previously described methods, to ensure all the relevant baseline characteristics were rebalanced within each subgroup. If post-weighting balance could not be reached in a certain subgroup, multivariable regression models were subsequently applied for covariate adjustments. We also conducted interaction tests between receipt of radiotherapy and these baseline characteristics to evaluate the heterogeneity of the prognostic impact of radiotherapy across the subgroups.

Absolute 5-year OS rates between non-radiotherapy and radiotherapy arms were analyzed by nonparametric sliding-window subpopulation treatment effect pattern plot (STEPP) analysis following procedure in a previous report [21–23]. Briefly, six parameters (age at diagnosis, pT category, HR status, ERBB2 status, cT category, and cN category) were chosen to construct a Cox proportional hazards model defining the composite risk. Then, the value of the composite risk of each patient was calculated by summing the model parameter estimates according to the observed prognostic factor values. Nonparametric sliding-window STEPP analysis was used to investigate the patterns of absolute treatment effect between the two arms, as measured by Kaplan-Meier estimates of 5-year OS (y-axis) across the continuum of values of composite risk (x-axis).

Furthermore, we conducted sensitivity analyses using multivariable Cox regressions with the aforementioned risk

Table 1 Baseline characteristics among the NCDB breast cancer patients diagnosed between 2010 and 2018 with cN+ disease who convert to ypN0 after NAC

Characteristic	Overall (N=3,351) No. (%)	No Radiotherapy (N=1,910)	Radiotherapy (N=1,441)	<i>P</i> ¹	Standardized Mean Difference	
					Unweighted	Weighted ²
Age				0.002	0.12	0.005
20–40	894 (26.7)	472 (24.7)	422 (29.3)			
40–60	1,894 (56.5)	1,089 (57.0)	805 (55.9)			
60+	563 (16.8)	349 (18.3)	214 (14.9)			
Ethnicity				0.21	0.07	0.010
Non-Hispanic White	2,207 (65.9)	1,267 (66.3)	940 (65.2)			
Non-Hispanic Black	572 (17.1)	335 (17.5)	237 (16.4)			
Hispanic	364 (10.9)	189 (9.9)	175 (12.1)			
Non-Hispanic Other	208 (6.2)	119 (6.2)	89 (6.2)			
pT category				0.011	0.10	0.004
pT0/IS	1,851 (55.2)	1,096 (57.4)	755 (52.4)			
pT1	1,208 (36.0)	649 (34.0)	559 (38.8)			
pT2	292 (8.7)	165 (8.6)	127 (8.8)			
Grade				0.14	0.07	0.004
Grade I	70 (2.1)	46 (2.4)	24 (1.7)			
Grade III	2,452 (73.2)	1,409 (73.8)	1,043 (72.4)			
Grade II	829 (24.7)	455 (23.8)	374 (26.0)			
Hormone receptor status				0.002	0.11	0.003
Negative	1,706 (50.9)	1,016 (53.2)	690 (47.9)			
Positive	1,645 (49.1)	894 (46.8)	751 (52.1)			
ERBB2 status				0.010	0.09	0.002
Negative	1,859 (55.5)	1,023 (53.6)	836 (58.0)			
Positive	1,492 (44.5)	887 (46.4)	605 (42.0)			
Histopathological classification				0.82	0.02	0.004
Ductal	3,051 (91.0)	1,744 (91.3)	1,307 (90.7)			
Lobular	203 (6.1)	113 (5.9)	90 (6.2)			
Other	97 (2.9)	53 (2.8)	44 (3.1)			
Charlson-Deyo score				0.26	0.06	0.004
0	3,005 (89.7)	1,700 (89.0)	1,305 (90.6)			
1	284 (8.5)	175 (9.2)	109 (7.6)			
≥2	62 (1.9)	35 (1.8)	27 (1.9)			
Diagnosis year				<0.001	0.14	0.007
2010–2012	624 (18.6)	323 (16.9)	301 (20.9)			
2013–2015	1,391 (41.5)	843 (44.1)	548 (38.0)			
2016–2018	1,336 (39.9)	744 (39.0)	592 (41.1)			
cT category				0.011	0.09	0.004
cT1	716 (21.4)	438 (22.9)	278 (19.3)			
cT2	2,635 (78.6)	1,472 (77.1)	1,163 (80.7)			
cN category				<0.001	0.41	0.003
cN1	2,840 (84.8%)	1,734 (90.8%)	1,106 (76.8%)			
cN2	296 (8.8%)	124 (6.5%)	172 (11.9%)			
cN3	215 (6.4%)	52 (2.7%)	163 (11.3%)			

¹Pearson's Chi-squared test²The weighted cohort indicated that assigning weights to each patient by inverse probability weighting

factors (i.e., adjustment for eleven baseline characteristics) to further confirm our findings.

All the analyses were performed by R version 4.2.3. And two-sided $P < .05$ was considered statistically significant. No adjustments were made for multiple comparisons.

Results

We identified 3,351 patients who met eligibility criteria, of whom 57.0% ($N=1,910$) did not receive radiotherapy and 43.0% ($N=1,441$) did receive radiotherapy (Fig. 1). Compared with non-radiotherapy patients, patients receiving

radiotherapy were more likely to be diagnosed at 20–40 years old (29.3% vs. 24.7%), have pT1-2 (47.6% vs. 42.6%), cT2 (80.7% vs. 77.1%), cN2-3 (23.2% vs. 9.2%), and ERBB2-negative status (58.0% vs. 53.6%) as shown in Table 1. Additionally, compared with Hispanic patients, of whom 48.0% received radiotherapy, non-Hispanic White (42.6%), non-Hispanic Black (41.4%) and non-Hispanic Other (42.8%) patients had lower proportions of receiving radiotherapy.

After IPW adjustment, SMDs were smaller than 0.1 for all baseline characteristics (Table 1 and Supplementary Figure S1) indicating that the weighted populations in the two subgroups were generally comparable.

The median follow-up time was 60.8 months (interquartile range [IQR] 44.3 to 81.2). Before and after IPW adjustment, no significant OS differences were observed between patients with and without radiotherapy (unweighted 5-year OS rate: 93.6% vs. 92.4%, log-rank $P = .25$; weighted 5-year OS rate: 92.7% vs. 92.6%, IPW-adjusted log-rank $P = .96$; IPW-adjusted hazard ratio = 1.01 [95% CI 0.77 to 1.32], $P = .96$; Fig. 2 and Supplementary Figure S2). Subgroup analyses based on cN category demonstrated that differences in OS rates between the two groups (with and without

RT) were significantly correlated with cN category, and an OS advantage was associated with RT receipt only among cN2-3 patients but not cN1 patients (cN1: hazard ratio = 1.26 [95% CI 0.94 to 1.70] vs. cN2-3: hazard ratio = 0.55 [95% CI 0.31 to 0.97]; P -interaction = 0.024; Fig. 3). The Kaplan-Meier curves for the cN1 and cN2-3 subgroups are shown in Fig. 4. No significant heterogeneity for the differences in OS between the two groups based on age, ethnicity, pT category, HR status, ERBB2 status, grade, and molecular subtype was observed (all P -interaction > 0.05; Fig. 3).

All 3,351 patients were included in the STEPP analysis, and the results defining the composite risk, thus validating the findings from the Cox model, are presented in Supplementary Table S1 and Figure S3. The STEPP analysis demonstrated no significant differences between the RT and no RT groups, regardless of the composite risk (Fig. 5).

In addition, the sensitivity analysis from multivariable Cox regressions with nine selected risk factors showed consistent results with the above IPW-adjusted results (hazard ratio = 1.05 [95% CI 0.79 to 1.39], $P = .74$).

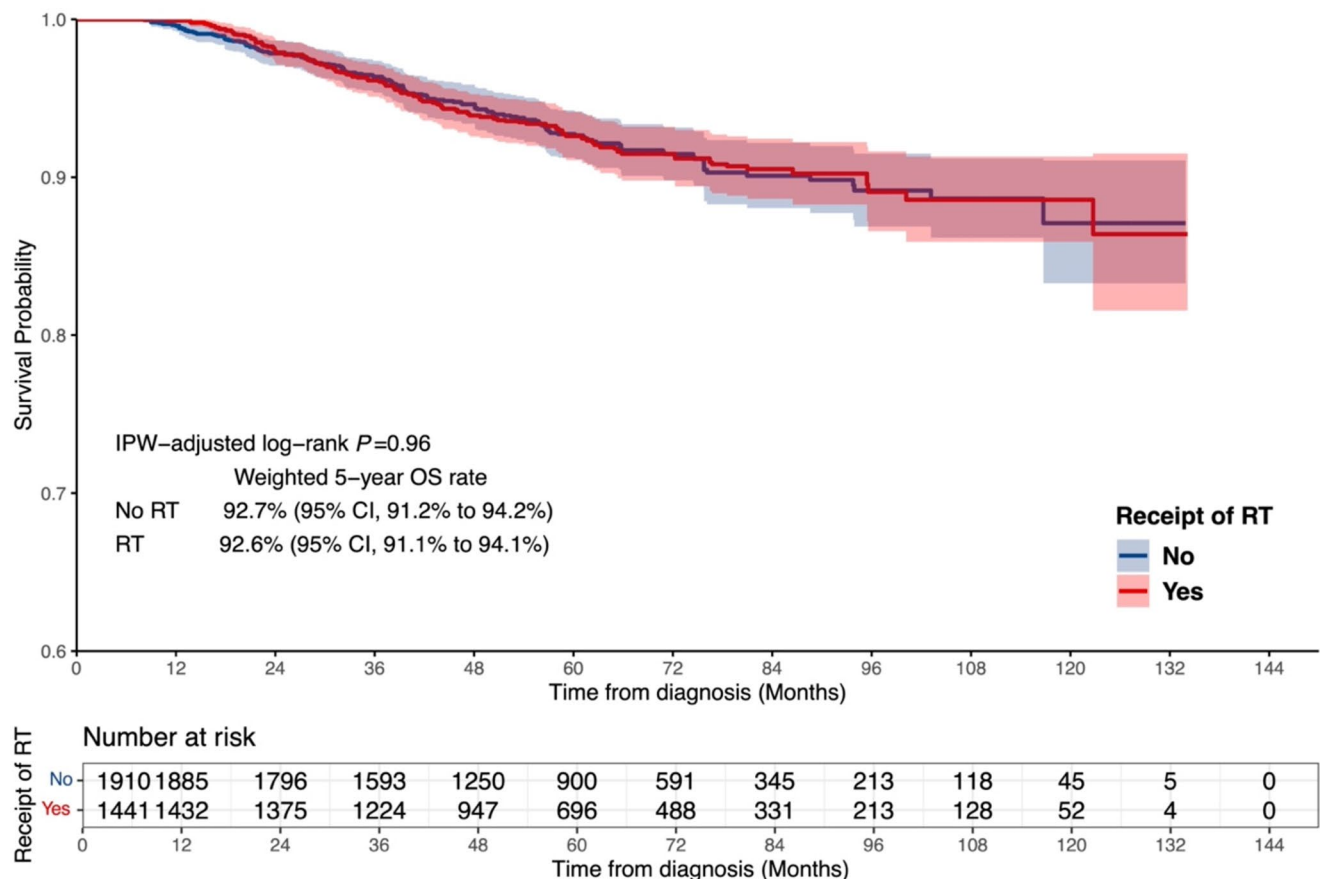


Fig. 2 Kaplan-Meier curves of radiotherapy versus no radiotherapy after IPW among the NCDB breast cancer patients diagnosed between 2010 and 2018 with cN+ disease who convert to ypN0 after NAC. Abbreviations: IPW, inverse probability weighting; OS, overall survival; RT, radiotherapy

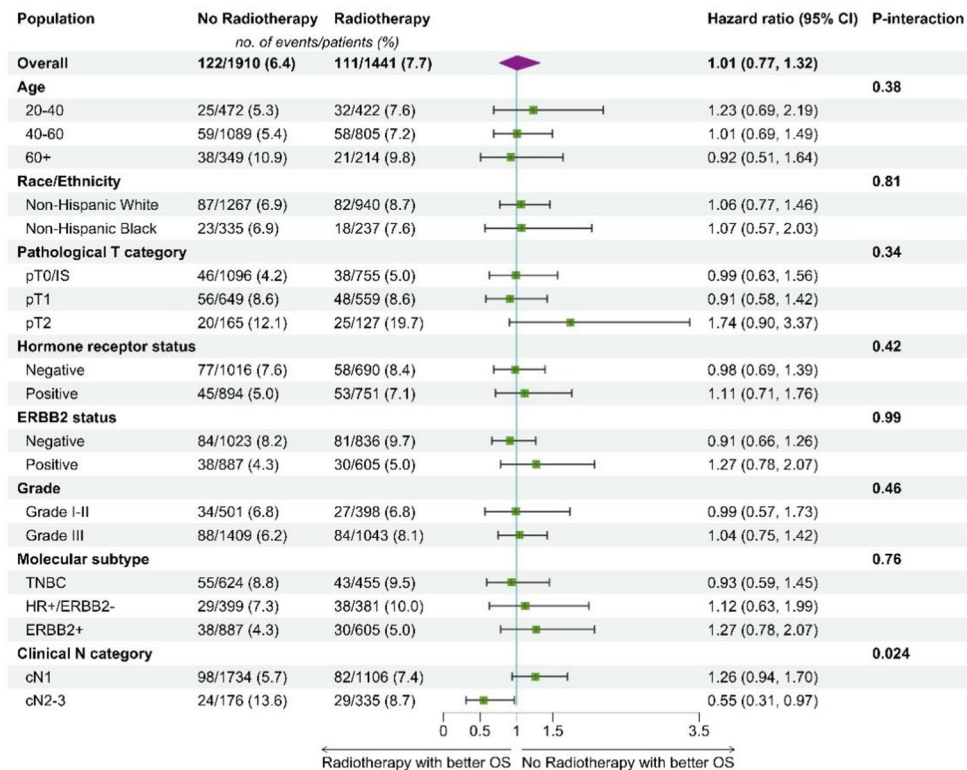


Fig. 3 Overall survival analyses based on multivariable Cox Proportional Hazards modeling by receipt of radiotherapy among the NCDB breast cancer patients diagnosed between 2010 and 2018 with cN+ disease who convert to ypN0 after NAC. Post-weighting balance was not achieved in the age 20–40, age 60+, non-Hispanic Black, pT1, pT2, HR-positive, grade I-II, TNBC, HR+/ERBB2- BC, cN2, and cN3 subgroups; therefore, multivariable Cox regression based on imputed datasets was instead applied. Other subgroup analyses still applied

Discussion

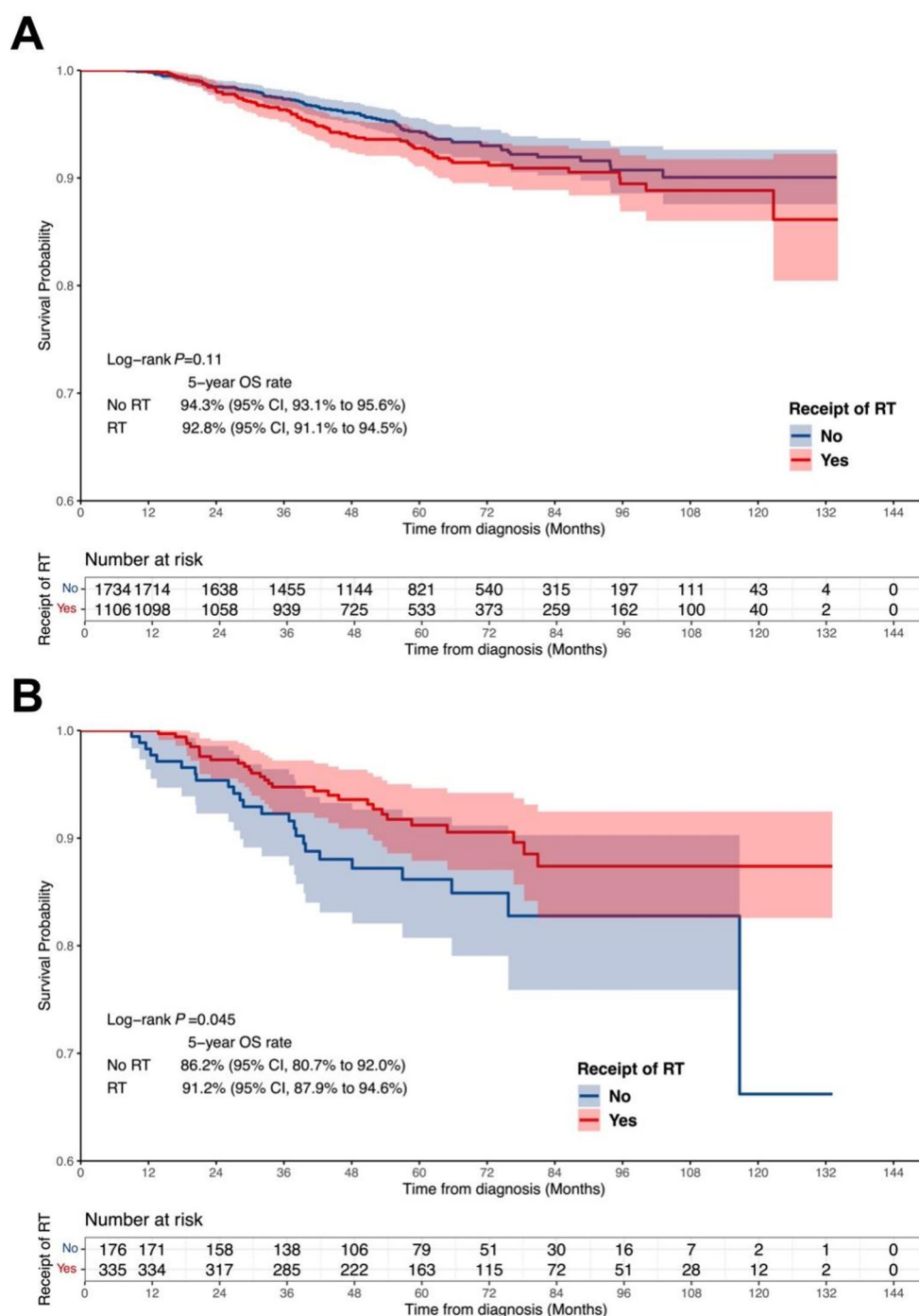
For patients with cN+ axillary disease that are down-staged to ypN0 at surgery, the controversy regarding the utility of PMRT after NAC persists. In this study, we evaluated the association between PMRT and survival among breast cancer patients with cN+ disease who achieved ypN0 after NAC, drawing on NCDB data from 2010 to 2018. Overall, we found no significant difference in OS between the RT and no RT groups. However, for patients with cN2-3 disease who converted to ypN0 after NAC, our findings suggest that radiotherapy may be potentially associated with improved OS, although further investigation is required.

In an era of taxane/anthracycline-based chemotherapy without targeted drugs for specific molecular subtypes, combined analysis of NSABP B18 and B27 demonstrated that patients with cN1 disease achieving pCR had a low LRR rate, suggesting that PMRT may provide minimal benefit in terms of long-term survival [11]. McGuire et al. also reported the 10-year LRR rate was zero for clinical stage I-II patients achieving both breast and axillary lymph node pCR

IPW method. P -interaction < 0.05, implied significant heterogeneity in the survival impact of receiving radiotherapy within subgroups of a specific variable. The multivariable Cox models were adjusted for age at diagnosis, ethnicity, pT category, tumor grade, HR status, ERBB2 status, histopathological classification, Charlson-Deyo score, diagnosis year, cT and cN category. Abbreviations: HR, hormone receptor; BC, breast cancer; TNBC, triple-negative breast cancer; NAC, neoadjuvant chemotherapy; NCDB, National Cancer Data Base

[12]. In the study by Scodan et al., they found that PMRT had no significant effect on the LRR or OS in patients with clinical stage II-III disease at diagnosis achieving pCR in both the breast and axillary lymph nodes [13]. However, a trend was shown towards worse OS among patients who did not achieve a pCR in the breast after NAC (HR 6.65; 95% CI 0.82 to 54.12; $P = .076$). Shim et al. reported a multicenter, retrospective study in Korea (KROG 12 – 05), finding that PMRT in ypN0 patients after NAC for clinical stage II-III breast cancer showed no correlation with differences in DFS or OS in multivariable analysis [14]. Recently, Huang et al. reported a multicenter, retrospective study of Chinese breast cancer patients with clinical T1-4N1-2M0 who underwent NAC and mastectomy. In 490 patients who were ypN0, 47% received PMRT. They also found no significant differences in DFS and OS between the RT and no RT groups before and after propensity score matching (PSM) [15]. A pooled analysis of two prospective trials focusing on the ERBB2-positive breast cancer subtype revealed that 55% of patients achieved nodal pCR with anti-ERBB2 targeted treatment, and those with axillary ypN0 disease had a 5-year

Fig. 4 Kaplan-Meier curves of overall survival for cN1 and cN2-3 subgroups. Panel **A** shows the Kaplan-Meier curves for the subgroup including patients with cN1 breast cancer. Panel **B** shows the Kaplan-Meier curves for the subgroup including patients with cN2 or cN3 breast cancer. Abbreviations: OS, overall survival

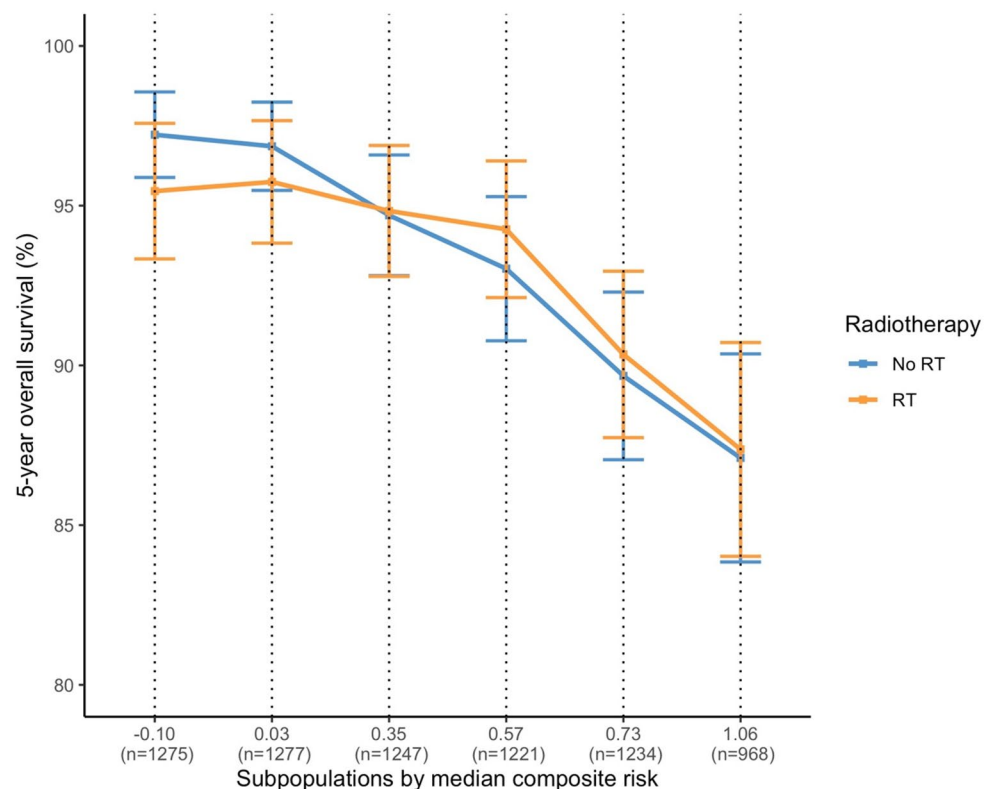


locoregional recurrence-free survival rate of 97%. PMRT had no effect on LRR or progression-free survival [16]. These findings collectively support the notion that PMRT may be reasonably considered for omission in patients with cN1 disease who achieve ypN0 status after NAC, as our IPW-based analysis also demonstrated no significant OS differences in this subgroup.

Another important randomized clinical trial, NSABP B51/ROG 1304, established high level of evidence that

supports PMRT omission in patients with cN1 disease [18, 19]. From 2013 to 2020, 1,641 patients with cT1-3N1M0 disease were randomized. After a median follow-up of 59.5 months, the estimated 5-year invasive breast cancer-free interval rate was 91.8% for the no RNI group and 92.7% for the RNI group (HR 0.88, 95%CI 0.60 to 1.29; $P = .60$). There was no significant difference in DFS and OS between the RNI and no RNI groups. However, this study did not specifically address the potential benefit of PMRT in

Fig. 5 Subpopulation treatment effect pattern plot (STEPP) of 5-year overall survival rates. The error bars indicate the 95% CI



patients with cN2-3 disease who achieve ypN0 status after NAC, which remains an area of controversy.

Evidence for cN2-3 patients was so far established on retrospective studies. McGuire et al. reported that PMRT reduced the LRR and improved the disease-specific survival and OS in stage III breast cancer, particularly in patients with cN2-N3 involvement [12]. Huang et al. reported that PMRT tended to improve DFS in cN2 patients who achieved ypN0 after NAC and mastectomy, but with no statistical significance (5-year DFS for No-PMRT vs. PMRT: 74.3% vs. 88.7%, $P = .086$) [15]. In this study, we observed a significant OS advantage associated with RT receipt in the cN2-3 subgroup with a sample size of 511 patients. Our STEPP analysis showed a similar trend in patients with a higher composite risk, although this was not significant. Overall, our findings suggest that PMRT may be a consideration for patients with more advanced nodal disease (cN2-3) who achieve ypN0 status after NAC, indicating that a more tailored approach to PMRT application may be warranted.

There are several limitations in this study. First, this is a retrospective cohort study based on data from the NCDB, which is a hospital-based tumor registry. Although it captures approximately 80% of all breast cancer diagnoses in the United States [24], it may not be fully representative of the entire breast cancer population, potentially introducing selection bias. Second, as an inherent limitation of the NCDB, we were unable to obtain detailed radiation therapy

parameters, including dose prescription, irradiated volumes, and delivery techniques. These are fundamental factors known to significantly influence treatment outcomes [25], and the lack of these data limits the robustness of our conclusions regarding the efficacy of PMRT. Third, the NCDB does not capture treatment-related toxicity data, which prevents us from weighing the potential survival benefits of PMRT against the risk of adverse effects, a critical consideration for clinical decision-making. Fourth, the sample size of cN2-3 patients is relatively small due to the rarity of this subgroup, and the upper bound of the 95% CI for the HR in this subgroup is 0.97, indicating that the observed potential benefit should be interpreted with caution.

In conclusion, this comprehensive analysis provides additional evidence consistent with the NSABP B51 findings, suggesting that PMRT may not be necessary for all ypN0 patients, particularly those initially diagnosed with cN1 disease. In comparison, PMRT may still be considered for those with initial cN2-3 disease who convert to ypN0, as there may be a potential survival benefit in this specific subgroup. These results indicate a potential need for a more tailored approach to the application of PMRT for patients with cN+ disease achieving ypN0 after NAC. However, due to the aforementioned limitations, our conclusions should be interpreted cautiously. Further prospective studies with larger cohorts are needed to refine guidelines for PMRT

use, ensuring optimal outcomes for all breast cancer patient subgroups.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10549-026-07981-x>.

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Data availability The data that support the findings of this study are available in National Cancer Database.

Declarations

Ethical approval The study was conducted in accordance with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. The data used in this study was obtained from the National Cancer Database (NCDB). The NCDB is a joint project of the American College of Surgeons and the American Cancer Society. The data is de-identified and aggregated, and its use for research purposes is in compliance with the Health Insurance Portability and Accountability Act (HIPAA) regulations. Ethical approval was not required for this study as the data does not contain any personally identifiable information. We have adhered to the highest ethical standards in the conduct of this research, including the responsible use and interpretation of the data.

Disclosures None.

Competing interests The authors declare no competing interests.

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